

Arrays In Java

11 August 2026 19:05

Java arrays are object that stores multiple elements of the **same data type** in a single contiguous memory block.

What is an Array?

An array is a collection of elements of the same type stored in a single contiguous block of memory. Arrays allow us to group similar data together and access them using an index.

Array in Java

In Java, an array is an object that stores elements of a similar data type in contiguous memory locations. It is a data structure used to store a fixed number of elements.

Arrays in Java are index based, meaning the first element is at index 0 and the second at index 1 and so on.

In Java, an array is considered an object of a dynamically created class.

Aries can store primitive values or objects and Java supports both single dimensional and multi-dimensional arrays.

Single dimensional Array

A single dimensional array in Java is a linear collection of elements of the same data type. It can be declared and instantiated using the following syntax:

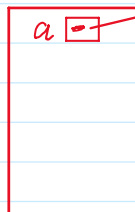
```
Datatype[] array_name; Or  
Datatype []array_name; Or  
Datatype array_name[];
```

Syntax of memory allocation of an Array:

```
Array_name = new Datatype[size];
```

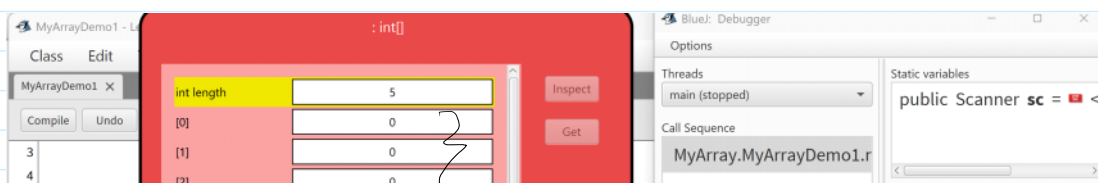
For example:

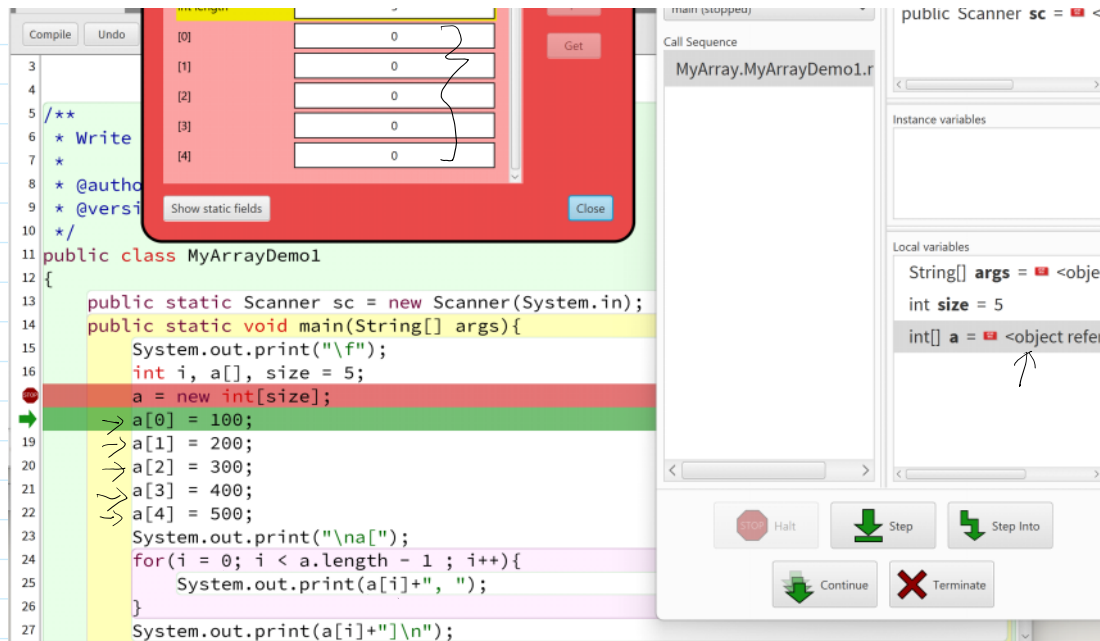
```
int []a;  
int size = 5;  
a = new int[size];
```



Heap

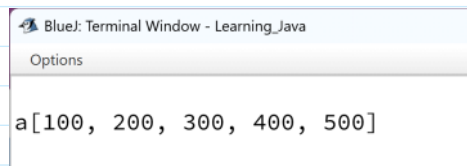
Keyword that allocates memory dynamically on Heap memory area.





All the values are inserted into each cell or variable.

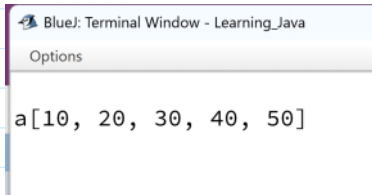
Output:



```
package MyArray;
import java.util.*;

/**
 * Write a description of class MyArrayDemo1 here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class MyArrayDemo1
{
    public static Scanner sc = new Scanner(System.in);
    public static void main(String[] args){
        System.out.print("\f");
        // Declaration of array along with initialization
        int a[] = {10, 20, 30, 40, 50};
        int i;
        System.out.print("\na[");
        for(i = 0; i < a.length - 1 ; i++){
            System.out.print(a[i]+", ");
        }
        System.out.print(a[i]+"]\n");
    }
}
```

Output:

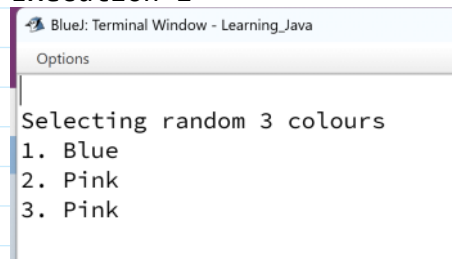


```
package MyArray;
import java.util.*;

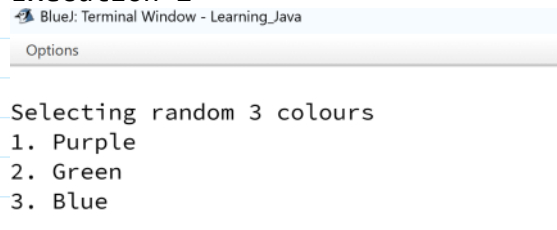
/**
 * Write a description of class MyArrayDemo2 here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class MyArrayDemo2
{
    public static Scanner sc = new Scanner(System.in);
    public static void main(String[] args){
        System.out.print("\f");
        String[] color = {"Red", "Green", "Blue", "Brown", "Purple",
"Pink"};
        Random r = new Random();
        System.out.print("\nSelecting random 3 colours");
        System.out.print("\n1. "+color[r.nextInt(color.length)]);
        System.out.print("\n2. "+color[r.nextInt(color.length)]);
        System.out.print("\n3. "+color[r.nextInt(color.length)]);
    }
}
```

Output:

Execution-1



Execution-2



```

package MyArray;
import java.util.*;

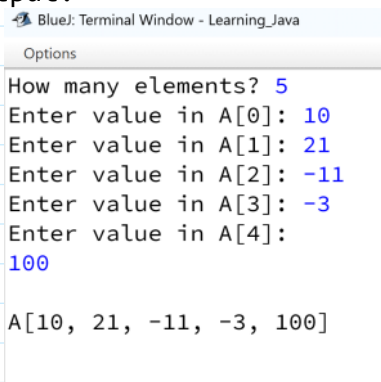
/**
 * Write a description of class InputAndOutputInSDA here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class InputAndOutputInSDA
{
    public static Scanner sc = new Scanner(System.in);
    public static void main(String[] args){
        System.out.print("\f");
        int i, A[], size;
        System.out.print("How many elements? ");
        size = sc.nextInt();
        A = new int[size];

        for(i = 0; i < size; i++){
            System.out.print("Enter value in A["+i+"]: ");
            A[i] = sc.nextInt();
        }

        System.out.print("\nA[");
        for(i = 0; i < size - 1; i++){
            System.out.print(A[i] + ", ");
        }
        System.out.print(A[i] + "]\n ");
    }
}

```

Output:



```

BlueJ: Terminal Window - Learning_Java
Options
How many elements? 5
Enter value in A[0]: 10
Enter value in A[1]: 21
Enter value in A[2]: -11
Enter value in A[3]: -3
Enter value in A[4]:
100

A[10, 21, -11, -3, 100]

```

WAP to create a SDA of size N, input values in the array and print the array. Find maximum value and minimum values from it.

```

package MyArray;
import java.util.*;

```

```

/**
 * Write a description of class InsertAtPosition here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class InsertAtPosition
{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        System.out.print("\nEnter number of elements: ");
        int n = sc.nextInt();
        //Array has one extra position for insertion.
        int[] arr = new int[n+1];
        System.out.print("Enter " + n + " elements: ");
        int i;
        for(i = 0; i < n; i++){
            arr[i] = sc.nextInt();
        }

        System.out.print("Enter element: ");
        int element = sc.nextInt();

        System.out.print("Enter position: ");
        int p = sc.nextInt();

        //validate position
        if(p < 0 || p > n){
            System.out.print("\nInvalid position.");
        }
        else{
            // Shift elements one position to the right.
            for( i = n; i > p; i--){
                arr[i] = arr[i-1];
            }

            //Insert the new element.
            arr[p] = element;
            System.out.print("\nArray after insertion: [");
            for(i = 0; i < n; i++){
                System.out.print(arr[i]+", ");
            }
            System.out.print(arr[i]+"]\n");
        }

        sc.close();
    }
}

```

Output:

Enter number of elements: 5

Enter 5 elements: 13

17

19

23

29

Enter element: 31

Enter position: 2

Array after insertion: [13, 17, 31, 19, 23, 29]